Comparison of prostate specific antigen test outcomes for prostate cancer diagnosis in different ethnic groups – the EPIC study

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Background

The Prostate Specific Antigen (PSA) blood test is a diagnostic tool for men with possible prostate cancer.

Average PSA levels vary for men of different ethnic groups.

Prostate cancer incidence and mortality in the UK are highest for Black men and lowest for Asian men.

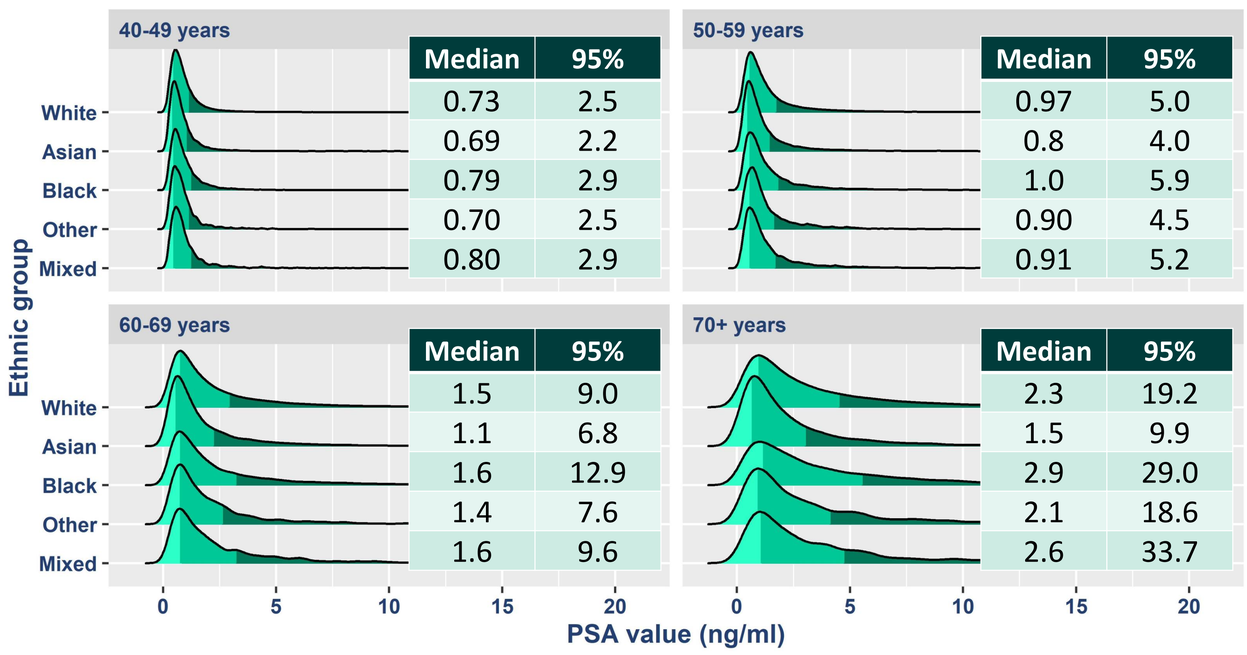
This study aimed to determine if there are differences in PSA performance between patients of different ethnic groups.

Table 1. Cohort characteristics

Ethnic group	Number of patients	Proportion of patients	Age-standardised prostate cancer rate/1000
White	649 445	88.9%	29.65
Asian	37 827	5.2%	12.15
Black	31 053	4.3%	45.23
Other	6 454	0.9%	22.03
Mixed	5 736	0.8%	32.05

Figure 1. PSA distribution by ethnic group and age category

Colour changes indicate 25% and 75% values, numerics for median and 95% values



Methods

Data source:

- Routine data from UK general practice, cancer registrations, and secondary care
- Supplied by the Clinical Practice Research Datalink (CPRD)

Included patients:

- Had a PSA test between 2010 and 2017
- Were aged at least 40 years old

Threshold for raised PSA:

- 40-49 year olds 2.5ng/ml
- 50-59 year olds 3.0ng/ml
- 60-69 year olds 4.0ng/ml
- 70 years and older 5.0ng/ml

Analysis:

- Multi-level logistic regression

Results

Black men had the highest prostate cancer incidence, at 45 cases/1000 men, with Asian men lowest at 12 cases/1000 men.

White men with a raised PSA were more likely to be diagnosed with prostate cancer than White men with a normal result. For Asian and Black men this statistic was much higher.

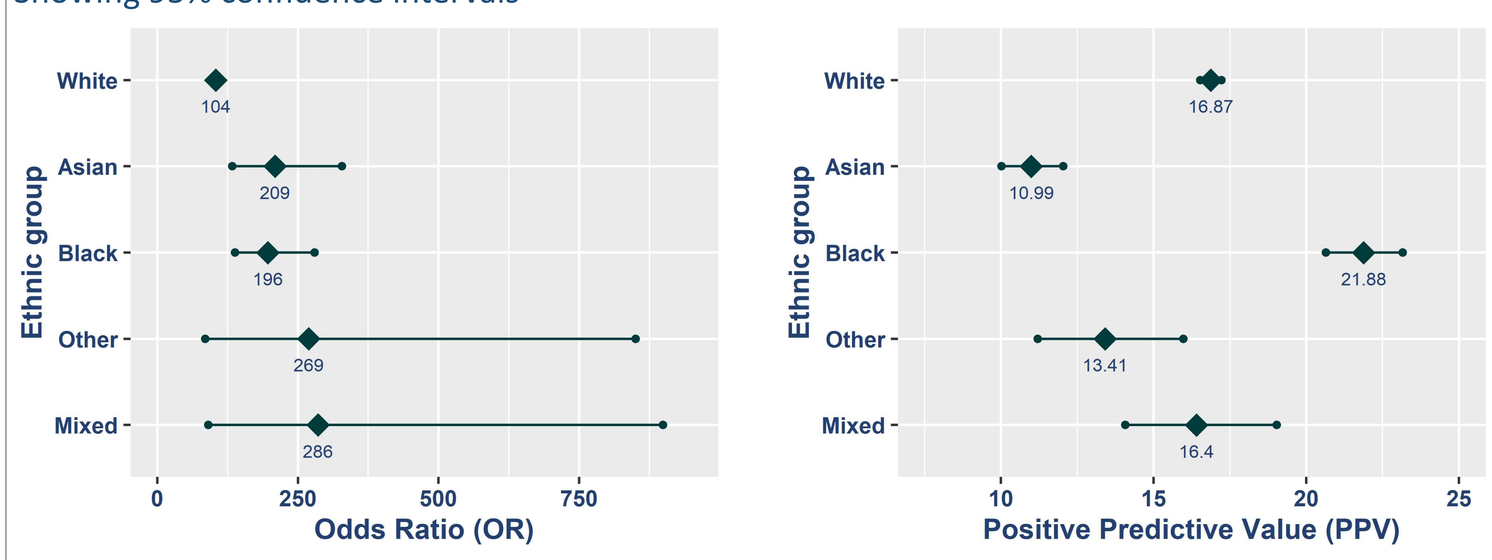
Results for the Mixed and Other ethnic groups were unclear, largely because there were fewer men in these categories.

A raised PSA result was much more likely to lead to a cancer diagnosis for Black men, with a PPV of 21.9, compared to Asian men who had the lowest PPV of 11.0.

Figure 2. PSA test performance in men from different ethnic groups

Adjusted for model covariates (general practice, age, deprivation, and year of test)

Showing 95% confidence intervals



Odds ratio for prostate cancer in men with a raised PSA result vs. men with a normal result

Positive Predictive Value of a raised PSA test for prostate cancer diagnosis

Conclusions

The predictive utility of PSA tests for prostate cancer differ according to ethnic group, being highest in Black men, and lowest for Asian men.

The strength of association of a raised test result to cancer diagnosis also varies by ethnicity, being around twice as strong for Asian and Black men relative to White men.

Further research is needed to better understand the causes of these differences and the clinical implications.

Acknowledgements

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Further information

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